KARTIK PATIL

AI AND DATA SCIENCE STUDENT

PROFESSIONAL SUMMARY

AI and Data Science Engineering student focused on building practical machine-learning applications, from explainable computer-vision models to voice and LLM-powered tools. Comfortable taking a project from a trained model to a working web interface, with an interest in open source and user-friendly design.

CONTACT

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PROJECTS

Leaf Scan AI (V1 & V2) | *Python, Keras, FastAPI, Grad-CAM*
Web app where users upload a leaf photo and a trained deep-learning model classifies it into 38 plant conditions (healthy and diseased) across 14 species. V2 uses a fine-tuned model and adds Grad-CAM heatmaps, so users can see which parts of the leaf drove each prediction.

Multilingual Language Model | *Python, Streamlit, OpenAI Whisper*
Voice-enabled assistant that lets users interact in 97 languages, using Whisper for speech recognition. API keys are kept out of the code through environment variables. Source code is on GitHub.

LearnMate AI | *React, FastAPI, Gemini, Firebase*
Adaptive learning agent built for a hackathon. Generates practice questions with Gemini and stores each student's history and topic mastery in Firebase.

Smart Railway Level-Crossing Safety System | *Arduino Uno, HC-SR04, Servo*
Uses two ultrasonic sensors to detect an approaching train and close a servo-driven gate automatically, with LED and buzzer warnings and live status on a 16x2 LCD.

Smart Classroom Occupancy & Energy Saver | *ESP32, PIR, DHT11, Relay*
Team IoT project that switches a fan and light through a relay based on room occupancy and monitors temperature and humidity to cut wasted energy.

EDUCATION

Nutan Maharashtra Institute of Engineering and Technology | AI and Data Science **2024 – 2028**

Adarsh Jr. College | HSC **2022 – 2024**

CES Prerna Secondary School | SSC **2022**

TECHNICAL SKILLS

Programming & ML
Python
Keras, CNNs, Computer Vision
Explainable AI (Grad-CAM)

Web & Backend
FastAPI, Streamlit, React
Firebase
OpenAI and Gemini APIs

Tools & Hardware
Git and GitHub
Arduino, ESP32
VS Code